✈️ Airline Reservation & Fleet Management System (ARFMS)

Language: ANSI C
Architecture: Structured Enterprise Console System
Storage: Binary File-Based Persistence
Version: 1.0

📌 Executive Summary

The Airline Reservation & Fleet Management System (ARFMS) is a structured aviation enterprise simulation platform developed entirely in ANSI C.

The system integrates core airline operational components including:

• Secure administrator authentication
• Flight route management
• Passenger booking system
• Real-time seat availability control
• Aircraft fleet management
• Revenue analytics
• Timestamp-based audit logging

ARFMS simulates centralized airline workflow using deterministic structured programming and binary file-based persistence without reliance on external databases.

🏗️ System Architecture

ARFMS follows a layered logical architecture:

• Authentication Layer
• Flight Management Module
• Booking Processing Engine
• Fleet Management Module
• Revenue Analytics Engine
• Audit Logging Layer
• Binary Persistence Layer

Although implemented in a single source file, functional responsibilities are logically separated.

🔐 Authentication Module

The system ensures secure administrative access.

Features:

• Automatic admin initialization on first execution
• Password hashing using djb2 algorithm
• No plaintext credential storage
• Binary verification against `airline_users.dat`

Default Credentials:

Username: admin
Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

🛬 Flight Management Module

Each flight record contains:

• Flight ID
• Origin
• Destination
• Total Seats
• Available Seats
• Ticket Price

Functional Capabilities:

• Add flight
• View flights
• Automatic seat availability initialization

Seat availability updates dynamically during booking operations.

Storage File: `flights.dat`

🎟️ Booking Processing Engine

Each booking record contains:

• Booking ID
• Flight ID
• Passenger Name
• Seats Booked
• Total Fare
• Timestamp

Operational Flow:

• Validate flight existence
• Check seat availability
• Deduct available seats
• Calculate fare
• Record transaction
• Persist booking

Fare Formula:

```
Total Fare = Seats Booked × Ticket Price
```

The system prevents overbooking through availability validation.

Storage File: `bookings.dat`

🛩️ Fleet Management Module

Each aircraft record contains:

• Aircraft ID
• Model
• Capacity
• Maintenance Cost

Functional Capability:

• Add aircraft record

Storage File: `aircrafts.dat`

This module enables fleet tracking and maintenance cost recording.

📊 Revenue Analytics Engine

The analytics module processes recorded booking transactions and computes:

• Mean (Average Fare)
• Variance
• Standard Deviation

Formulas Used:

```
Mean = Σ Fare / N
Variance = Σ (Fare − Mean)² / N
Standard Deviation = √Variance
```

Purpose:

• Evaluate revenue distribution
• Measure booking variability
• Analyze airline financial performance patterns

📝 Audit Logging

All major system activities are recorded in:

`airline_logs.txt`

Logged Events:

• Admin login
• Flight addition
• Booking processing
• Aircraft addition
• Analytics execution
• System exit

Each log entry includes timestamp metadata for traceability.

💾 Data Storage Design

Binary files used:

File	Purpose
airline_users.dat	Authentication records
flights.dat	Flight database
bookings.dat	Booking transactions

File	Purpose
aircrafts.dat	Fleet records
airline_logs.txt	Audit logs

Binary storage ensures:

• Efficient record alignment
• Faster read/write operations
• Persistent aviation data integrity
• Reduced storage overhead

⚙️ Compilation & Execution

Compile:

```
gcc airline_management_system.c -o airline -lm
```

Run:

```
./airline
```

🧠 Engineering Highlights

• ANSI C structured programming
• File-based aviation database simulation
• Real-time seat inventory control
• Revenue computation logic
• Statistical financial analysis
• Deterministic memory management
• Timestamp-based operational traceability

🎯 Project Significance

ARFMS demonstrates the capability to design and implement centralized airline operational workflow using low-level systems programming.

It reflects:

• Aviation process modeling
• Structured data persistence
• Revenue analysis integration
• Secure authentication handling
• Enterprise-grade booking logic

🏁 Conclusion

The Airline Reservation & Fleet Management System (ARFMS) is a complete aviation enterprise simulation platform implemented in pure ANSI C.

It integrates:

Authentication
Flight Management
Booking Processing
Fleet Tracking
Revenue Analytics
Audit Logging

All within a structured and deterministic runtime architecture.